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Assginments:13.4

Task:01

Refactoring Data Transformation

Logic

Scenario

You are maintaining a data preprocessing script where numerical transformations are written using verbose loops.

Task Description

- Review the legacy code that computes transformed values using an explicit loop.
- Use an AI tool to suggest a more Pythonic refactoring approach.
- Refactor the code using list comprehensions or helper functions while preserving the output. Legacy Code

values = [2, 4, 6, 8, 10]

doubled = [] for v in values:

doubled.append(v * 2) print(doubled)

Expected Output

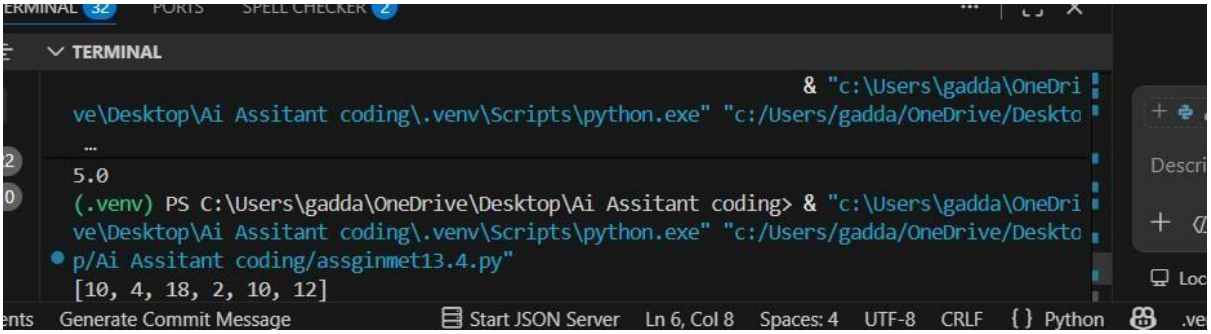
[4, 8, 12, 16, 20]

Prompt: for this code remove the unwanted lines for code refactoring

Code:

```
ssginmet13.4.py > ...
#task 1
values=[5, 2, 9, 1, 5, 6]
s=[value*2 for value in values]
print(s)
```

Output:



Observation:

The legacy code uses a **for loop with append()** to manually double each element in the list.

This approach works correctly but is **verbose and less Pythonic** for simple transformations.

It can be refactored using a **list comprehension** to make the code shorter, cleaner, and more readable while producing the same output [4, 8, 12, 16, 20]

Task:02

Improving Text Processing Code

Readability

Scenario

You are working on a log message generator that builds messages word by word.

Task Description

- Analyze the legacy code that constructs a sentence using repeated string concatenation.
- Ask AI to suggest a more efficient and readable approach.
- Refactor the code accordingly while keeping the final output unchanged. Legacy Code words = ["Refactoring", "with", "AI", "improves",

"quality"] message = ""

for w in words:

message += w + " "

print(message.strip())

Expected Output

Refactoring with AI improves quality

Prompt:for this code make it the code readble by applying the code refactoring Code:

```
#task02
words = ["Refactoring", "with", "AI", "improves","quality"]
concat= " ".join(words)
print(concat)
```

Output:

```
OUTPUT   DEBUG CONSOLE   TERMINAL 32   PORTS   SPELL CHECKER 2
PROBLEMS                 TERMINAL
Filter (e.g. text, **/*.ts, !**/...
assginment12.4.py  22
(.venv) PS C:\Users\gadda\OneDrive\Desktop\Ai Assitant coding> & "c:\Users\gadda\OneDrive\Desktop\Ai Assitant coding\.venv\Scripts\python.exe" "c:/Users/gadda/OneDrive/Desktop/Ai Assitant coding/assginmet13.4.py"
Refactoring with AI improves quality
```

Observation:

1. The legacy code builds the sentence using **repeated string concatenation inside a loop**.
2. This approach is **less efficient and less readable**, especially for longer lists of words.
3. It can be refactored using **" ".join(words)**, which is more Pythonic, efficient, and produces the same output: *"Refactoring with AI improves quality"*.

Task:03

Safer Access to Configuration Data

Scenario

You are maintaining a configuration loader where dictionary keys may or may not exist depending on deployment settings.

Task Description

- Review the legacy code that manually checks for dictionary keys. • Use AI suggestions to refactor the code using safer dictionary access methods.
- Ensure the behavior remains the same for missing keys. Legacy Code config = {"host": "localhost", "port": 8080} if "timeout" in config: print(config["timeout"]) else: print("Default timeout used") Expected Output Default timeout used Prompt:

For this legacy code remove unwanted lines of code give me code readable by code refactor Code:

```
#task03
config = {"host": "localhost", "port": 8080}
print(config.get("timeout", "Default timeout used"))
```

Output:

```
Filter (e.g. text, **/*.ts, !**/...
> assginment12.4.py 22
> assginmet13.4.py 10

(.venv) PS C:\Users\gadda\OneDrive\Desktop\Ai Assitant coding> & "c:\Users\gadda\OneDri
ve\Desktop\Ai Assitant coding\.venv\Scripts\python.exe" "c:/Users/gadda/OneDrive/Deskto
p/Ai Assitant coding/assginmet13.4.py"
Refactoring with AI improves quality
Default timeout used
```

Observation

The legacy code **manually checks if the key exists in the dictionary** using if "timeout" in config.

This approach works but makes the code **longer and less concise** for simple key access.

it can be refactored using **config.get("timeout", "Default timeout used")** for safer and more Pythonic dictionary access while keeping the same behavior.

Task:04

Scenario

You are enhancing a utility module that performs different operations based on user input.

Task Description

• Examine the multiple if–elif conditions used to determine operations.

• Ask AI to suggest a cleaner, scalable alternative.

• Refactor the logic using mapping techniques while preserving

functionality. Legacy Code action = "divide" x, y = 10, 2 if action == "add":

result = x + y elif action

== "subtract": result = x -

y elif action == "multiply":

result = x * y elif action

== "divide": result = x / y

else: result = None

print(result)

Expected Output

5.0

Prompt:for this code perform the division without performing any operation Code:

```
import operator

operations = {
    "add": operator.add,
    "subtract": operator.sub,
    "multiply": operator.mul,
    "divide": operator.truediv
}

action = "divide"
x, y = 10, 2
result = operations.get(action, lambda a, b: None)(x, y)
print(result)
#task05
```

Output:

```
> assginmet13.4.py  10
Default timeout used
5.0
Item Available
```

Observation:

The legacy code uses **multiple if–elif conditions** to choose the operation based on the action value.

This approach works but becomes **harder to maintain and scale** when more operations are added.

It can be refactored using a **dictionary mapping of actions to functions/operations**, making the code cleaner, more readable, and easier to extend while producing the same result 5.0

Task:05

Simplifying Search Logic in

Collections

Scenario

You are reviewing legacy inventory-check code that uses manual loops to find

elements.

Task Description

• Identify the explicit loop used for searching an item.

• Use AI assistance to refactor the logic into a more concise and readable form.

• Maintain the same output behavior. Legacy Code

inventory = ["pen", "notebook", "eraser", "marker"]

found = False for item in inventory: if item == "eraser":

found = True break

print("Item Available" if found else "Item Not Available")

Expected Output

Item Available

Prompt: Refactor the Python code that manually searches for an item in a list using a loop and a flag variable. Suggest a more concise and Pythonic approach while keeping the same output." Code:

```python
(variable) inventory: list[str]
#task05
inventory = ["pen", "notebook", "eraser", "marker"]

found = "eraser" in inventory

print("Item Available" if found else "Item Not Available")
```

Output:

```
assginment12.4.py 22
assginmet13.4.py 10
Refactoring with AI improves quality
Default timeout used
5.0
Item Available
(.venv) PS C:\Users\gadda\OneDrive\Desktop\Ai Assitant coding>
```

Observation

The legacy code uses a **manual loop with a flag variable (found)** to search for an item in the list.

This approach works but is **longer and less Pythonic** for simple membership checks.

It can be simplified using the **in operator**, which directly checks if "eraser" exists in the list while keeping the same output.